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Modeling and Discrete Simulation

Subject 8

Input Modeling

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Purpose & Overview

- Input models provide the driving force for a simulation model.
- The quality of the output is no better than the quality of inputs.
- In this lecture, we will discuss the 4 steps of input model development:
  - Collect data from the real system
  - Identify a probability distribution to represent the input process
  - Choose parameters for the distribution
  - Evaluate the chosen distribution and parameters for goodness of fit.

Slides are derived from the course textbook "Banks, Carson, Nelson, Nicol. Discrete-Event System Simulation, 5th Edition, Prentice Hall, 2010" and from textbook's course slides.

Contents

- Identifying the distribution
- Histograms
- Selecting families of distribution
- Parameter estimation
- Goodness-of-fit tests
- Fitting a non-stationary process

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IDENTIFYING THE DISTRIBUTION

Data Collection

- One of the biggest tasks in solving a real problem.
- Before “fitting”, accurate and relevant data must be available or collectible.
- Suggestions that may enhance and facilitate data collection:
  - Plan ahead: begin by a practice or pre-observing session, watch for unusual circumstances
  - Analyze the data as it is being collected: check adequacy
  - Combine homogeneous data sets, e.g. successive time periods, or during the same time period on successive days
  - Be aware of data censoring: the quantity is not observed in its entirety, danger of leaving out long process times
  - Check for relationship between variables, e.g. build scatter diagram
  - Check for autocorrelation
  - Collect input data, not performance data

Identifying the Distribution

- Histograms
- Selecting families of distribution
- Parameter estimation
- Goodness-of-fit tests
- Fitting a non-stationary process

Histograms

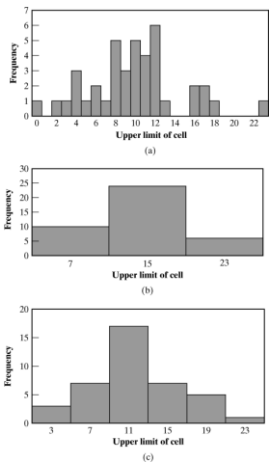
- A frequency distribution or histogram is useful in determining the shape of a distribution
- The number of class intervals depends on:
  - The number of observations
  - The dispersion of the data
  - Suggested: the square root of the sample size
- For continuous data:
  - Corresponds to the probability density function of a theoretical distribution
- For discrete data:
  - Corresponds to the probability mass function
  - If few data points are available: combine adjacent cells to eliminate the ragged appearance of the histogram

Histograms

Original Data
– **too ragged**

Combining adjacent cells
– **too coarse**

Combining adjacent cells
– **appropriate**

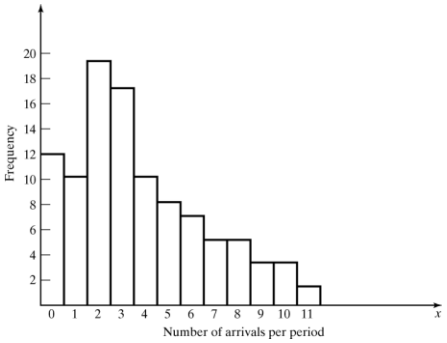


Same data with different interval sizes

Histograms

- Vehicle Arrival Example: # of vehicles arriving at an intersection between 7 am and 7:05 am was monitored for *100* random workdays.

Arrivals per Period	Frequency
0	12
1	10
2	19
3	17
4	10
5	8
6	7
7	5
8	5
9	3
10	3
11	1



- There are ample data, so the histogram may have a cell for each possible value in the data range

Selecting the Family of Distributions

- The purpose of preparing a histogram is to infer a known ***pdf*** or ***pmf***.
- A family of distributions is selected based on:
  - The context of the input variable
  - Shape of the histogram
- Frequently encountered distributions:
  - Easier to analyze: exponential, normal and Poisson
  - Harder to analyze: beta, gamma and Weibull

Selecting the Family of Distributions

- Use the physical basis of the distribution as a guide, for example:
  - Binomial: models the # of successes in *n* trials
  - Poisson: models the # of independent events that occur in a fixed amount of time or space
  - Normal: models the distribution of a process that is the sum of a number of component processes
  - Exponential: models the time between independent events, or a process time that is memoryless
  - Weibull: models the time to failure for components
  - Discrete or continuous uniform: models complete uncertainty
  - Triangular: models the a process for which only the minimum, most likely, and maximum values are known
  - Empirical: resamples from the actual data collected

Selecting the Family of Distributions

- Remember the physical characteristics of the process
  - Is the process naturally discrete or continuous valued?
  - Is it bounded?
- No "true" distribution for any stochastic input process
- Goal: obtain a good approximation

Quantile-Quantile Plots

- *Q-Q* plot is a useful tool for evaluating distribution fit
- If X is a random variable with ***cdf*** F, then the q-quantile of X is the γ such that

$$F(\gamma) = P(X \le \gamma) = q, \qquad \text{for } 0 < q < 1$$

  - When F has an inverse, $\gamma = F^{-1}(q)$

- Let $\{x_i, i = 1,2, \ldots, n\}$ be a sample of data from X and $\{y_j, j = 1,2, \ldots, n\}$ be the observations in ascending order:

$$y_j \text{ is approximately } F^{-1}\left(\frac{j - 0.5}{n}\right)$$

  where j is the ranking or order number

Quantile-Quantile Plots

- The plot of y_j versus $F^{-1}((j-0.5)/n)$ is
  - Approximately **a straight line** if F is a member of an appropriate family of distributions
  - The line has **slope 1** if F is a member of an appropriate family of distributions with appropriate parameter values

Quantile-Quantile Plots

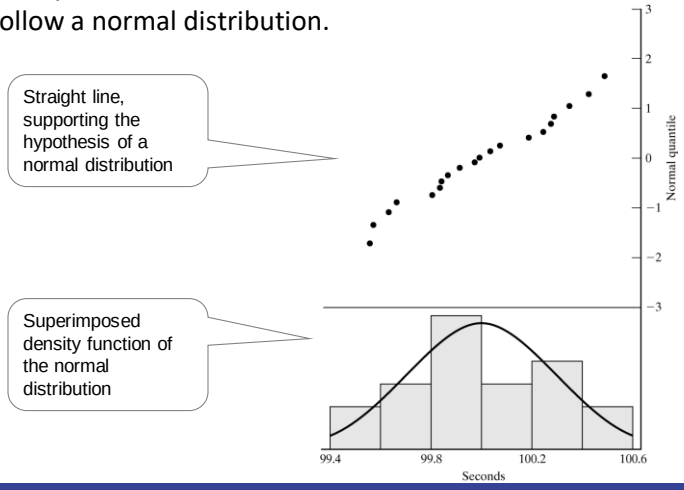
- Example: Check whether the door installation times follows a normal distribution.
  - The observations are now ordered from smallest to largest:

j	Value	j	Value	j	Value
1	99.55	6	99.98	11	100.26
2	99.56	7	100.02	12	100.27
3	99.62	8	100.06	13	100.33
4	99.65	9	100.17	14	100.41
5	99.79	10	100.23	15	100.47

  - y_j are plotted versus $F^{-1}((j-0.5)/n)$ where F has a normal distribution with the sample mean *(99.99 sec)* and sample variance (0.2832^2 sec^2)

Quantile-Quantile Plots

- Example (continued): Check whether the door installation times follow a normal distribution.



Quantile-Quantile Plots

- Consider the following while evaluating the linearity of a *q-q* plot:
  - The observed values never fall exactly on a straight line
  - The ordered values are ranked and hence not independent, unlikely for the points to be scattered about the line
  - Variance of the extremes is higher than the middle. Linearity of the points in the middle of the plot is more important.
- *Q-Q* plot can also be used to check homogeneity
  - Check whether a single distribution can represent both sample sets
  - Plotting the order values of the two data samples against each other

Parameter Estimation

- Next step after selecting a family of distributions
- If observations in a sample of size n are $X_1, X_2, \ldots, X_n$ (discrete or continuous), the sample mean and variance are:

$$\bar{X} = \frac{\sum_{i=1}^{n} X_i}{n} \qquad S^2 = \frac{\sum_{i=1}^{n} X_i^2 - n\bar{X}^2}{n-1}$$

- If the data are discrete and have been grouped in a frequency distribution:

$$\bar{X} = \frac{\sum_{j=1}^{n} f_j X_j}{n} \qquad S^2 = \frac{\sum_{j=1}^{n} f_j X_j^2 - n\bar{X}^2}{n-1}$$

where f_j is the observed frequency of value X_j

Parameter Estimation

- When raw data are unavailable (data are grouped into class intervals), the approximate sample mean and variance are:

$$\bar{X} = \frac{\sum_{j=1}^{c} f_j m_j}{n} \qquad S^2 = \frac{\sum_{j=1}^{n} f_j m_j^2 - n\bar{X}^2}{n-1}$$

where f_j is the observed frequency of in the jth class interval
m_j is the midpoint of the jth interval, and c is the number of class intervals

- A parameter is an unknown constant, but an estimator is a statistic.

Parameter Estimation

- Vehicle Arrival Example: # of vehicles arriving at an intersection between 7 am and 7:05 am was monitored for *100* random workdays.

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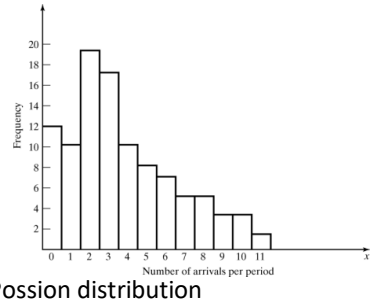
Parameter Estimation

- Vehicle Arrival Example (continued): Table in the histogram example on slide 8 (Table 9.1 in textbook) can be analyzed to obtain:

$$n = 100,\ f_1 = 12,\ X_1 = 0,\ f_2 = 10,\ X_2 = 1, \ldots,$$
$$\text{and } \sum_{j=1}^{k} f_j X_j = 364, \text{ and } \sum_{j=1}^{k} f_j X_j^2 = 2080$$

  - The sample mean and variance are

$$\bar{X} = \frac{364}{100} = 3.64$$
$$S^2 = \frac{2080 - 100*(3.64)^2}{99}$$
$$= 7.63$$



  - The histogram suggests *X* to have a Possion distribution
    - However, note that sample mean is not equal to sample variance.
    - Reason: each estimator is a random variable, is not perfect.

Parameter Estimation

Distribution	*Parameter(s)*	*Suggested Estimator(s)*
Poisson	α	$\hat{\alpha} = \bar{X}$
Exponential	λ	$\hat{\lambda} = \frac{1}{\bar{X}}$
Gamma	β, θ	$\hat{\beta}$ (see Table A.9)
		$\hat{\theta} = \frac{1}{\bar{X}}$
Normal	μ, σ^2	$\hat{\mu} = \bar{X}$
		$\hat{\sigma}^2 = S^2$ (unbiased)
Lognormal	μ, σ^2	$\hat{\mu} = \bar{X}$ (after taking ln of the data)
		$\hat{\sigma}^2 = S^2$ (after taking ln of the data)

Parameter Estimation

Weibull with $\nu = 0$	α, β	$\hat{\beta}_0 = \frac{\bar{X}}{S}$ $\hat{\beta}_j = \hat{\beta}_{j-1} - \frac{f(\hat{\beta}_{j-1})}{f'(\hat{\beta}_{j-1})}$ See Equations (9.12) and (9.15) for $f(\hat{\beta})$ and $f'(\hat{\beta})$ Iterate until convergence $\hat{\alpha} = \left(\frac{1}{n}\sum_{i=1}^{n} X_i^{\hat{\beta}}\right)^{1/\hat{\beta}}$
Beta	β_1, β_2	$\Psi(\hat{\beta}_1) + \Psi(\hat{\beta}_1 - \hat{\beta}_2) = \ln(G_1)$ $\Psi(\hat{\beta}_2) + \Psi(\hat{\beta}_1 - \hat{\beta}_2) = \ln(G_2)$ where Ψ is the digamma function, $G_1 = \left(\prod_{i=1}^{n} X_i\right)^{1/n}$ and $G_2 = \left(\prod_{i=1}^{n} (1 - X_i)\right)^{1/n}$

GOODNESS-OF-FIT TESTS

Goodness-of-Fit Tests

- Conduct hypothesis testing on input data distribution using:
  - Kolmogorov-Smirnov test
  - Chi-square test
- No single correct distribution in a real application exists.
  - If very little data are available, it is unlikely to reject any candidate distributions
  - If a lot of data are available, it is likely to reject all candidate distributions

Chi-Square test

- **Intuition**: comparing the histogram of the data to the shape of the candidate density or mass function
- Valid for large sample sizes when parameters are estimated by maximum likelihood
- By arranging the *n* observations into a set of *k* class intervals or cells, the test statistics is:

$$\chi_0^2 = \sum_{i=1}^{k} \frac{(O_i - E_i)^2}{E_i}$$

Observed Frequency

Expected Frequency
$E_i = n*p_i$
where p_i is the theoretical prob. of the *i*th interval.
Suggested Minimum = 5

which **approximately** follows the chi-square distribution with *k-s-1* degrees of freedom, where **s = # of parameters** of the hypothesized distribution estimated by the sample statistics.

Chi-Square test

- The hypothesis of a chi-square test is:

  H_0: The random variable, X, conforms to the distributional assumption with the parameter(s) given by the estimate(s).

  H_a: The random variable X does not conform.

- If the distribution tested is discrete and combining adjacent cell is not required (so that $E_i >$ minimum requirement):
  - Each value of the random variable should be a class interval, unless combining is necessary, and

$$p_i = p(x_i) = P(X = x_i)$$

Chi-Square test

- If the distribution tested is continuous:

$$p_i = \int_{a_{i-1}}^{a_i} f(x)\,dx = F(a_i) - F(a_{i-1})$$

  where a_{i-1} and a_i are the endpoints of the ith class interval and $f(x)$ is the assumed pdf, $F(x)$ is the assumed cdf.
  - Recommended number of class intervals (k):

Sample Size, n	Number of Class Intervals, k
20	Do not use the chi-square test
50	5 to 10
100	10 to 20
> 100	$n^{1/2}$ to n/5

  - Caution: Different grouping of data (i.e., k) can affect the hypothesis testing result.

Chi-Square test

- Vehicle Arrival Example (continued):

  H_0: the random variable is Poisson distributed.

  H_a: the random variable is not Poisson distributed.

1st calculate the mean
2nd find p_i
3rd find E_i
4th do the test

$$E_i = np(x) = n\frac{e^{-\alpha}\alpha^x}{x!}$$

x_i	Observed Frequency, O_i	Expected Frequency, E_i	$(O_i - E_i)^2/E_i$
0	12	2.6	7.87
1	10	9.6	
2	19	17.4	0.15
3	17	21.1	0.8
4	19	19.2	4.41
5	6	14.0	2.57
6	7	8.5	0.26
7	5	4.4	11.62
8	5	2.0	
9	3	0.8	
10	3	0.3	
> 11	1	0.1	
	100	100.0	27.68

p(0) = 0.026
p(1) = 0.096
p(2) = 0.174
p(3) = 0.211
p(4) = 0.192
p(5) = 0.140
p(6) = 0.085
p(7) = 0.044
p(8) = 0.020
p(9) = 0.008
p(10) = 0.003
p(≥11) = 0.001

Combined because of min E_i

– Degree of freedom is *k-s-1 = 7-1-1 = 5*, hence, the hypothesis is rejected at the *0.05* level of significance.

$$\chi_0^2 = 27.68 > \chi^2_{0.05,5} = 11.07$$

Kolmogorov-Smirnov Test

- Intuition: formalize the idea behind examining a *q-q* plot
- Recall from previous lectures:
  - The test compares the continuous cdf, *F(x)*, of the hypothesized distribution with the empirical cdf, $S_N(x)$, of the *N* sample observations.
  - Based on the maximum difference statistics (Tabulated in A.8):

    $D = max| F(x) - S_N(x)|$
- A more powerful test, particularly useful when:
  - Sample sizes are small,
  - No parameters are estimated from the data to construct the empirical cdf $S_N(x)$. *(In chi-square we need to specify the class intervals)*
- When parameter estimates have been made:
  - Critical values in Table A.8 (Kolmogorov-Smirnov Critical Values Table) are biased, too large.
  - More conservative, i.e., smaller Type I error than specified.

Kolmogorov-Smirnov Test

Suppose that 50 interarrival times (in minutes) are collected over the following 100-minute interval (arranged in order of occurrence):

0.44	0.53	2.04	2.74	2.00	0.30	2.54	0.52	2.02	1.89	1.53	0.21
2.80	0.04	1.35	8.32	2.34	1.95	0.10	1.42	0.46	0.07	1.09	0.76
5.55	3.93	1.07	2.26	2.88	0.67	1.12	0.26	4.57	5.37	0.12	3.19
1.63	1.46	1.08	2.06	0.85	0.83	2.44	1.02	2.24	2.11	3.15	2.90
6.58	0.64										

The null hypothesis and its alternate are formed as follows:

H_0: the interarrival times are exponentially distributed.
H_1: the interarrival times are not exponentially distributed.

The data were collected over the interval from 0 to $T = 100$ minutes. It can be shown that, if the underlying distribution of interarrival times $\{T_1, T_2, \ldots\}$ is exponential, the arrival times are uniformly distributed on the interval $(0, T)$. The arrival times $T_1, T_1 + T_2, T_1 + T_2 + T_3, \ldots, T_1 + \cdots + T_{50}$ are obtained by adding interarrival times. The arrival times are then normalized to a (0, 1) interval so that the Kolmogorov–Smirnov test, as presented in Section 7.4.1, can be applied. On a (0, 1) interval, the points will be $[T_1/T, (T_1 + T_2)/T, \ldots, (T_1 + \cdots + T_{50})/T]$. The resulting 50 data points are as follows:

Kolmogorov-Smirnov Test

0.0044	0.0097	0.0301	0.0575	0.0775	0.0805	0.1059	0.1111	0.1313	0.1502
0.1655	0.1676	0.1956	0.1960	0.2095	0.2927	0.3161	0.3356	0.3366	0.3508
0.3553	0.3561	0.3670	0.3746	0.4300	0.4694	0.4796	0.5027	0.5315	0.5382
0.5494	0.5520	0.5977	0.6514	0.6526	0.6845	0.7008	0.7154	0.7262	0.7468
0.7553	0.7636	0.7880	0.7982	0.8206	0.8417	0.8732	0.9022	0.9680	0.9744

Following the procedure in Example 7.6 produces a D^+ of 0.1054 and a D^- of 0.0080. Therefore, the Kolmogorov–Smirnov statistic is $D = \max(0.1054, 0.0080) = 0.1054$. The critical value of D obtained from Table A.8 for a level of significance of $\alpha = 0.05$ and $n = 50$ is $D_{0.05} = 1.36/\sqrt{n} = 0.1923$; but $D = 0.1054$, so the hypothesis that the interarrival times are exponentially distributed cannot be rejected.

p-Values and "Best Fits"

- *p-value* for the test statistics
  - The significance level at which one would just reject H_0 for the given test statistic value.
  - It represents the smallest level of significance that would lead to rejection of $\boldsymbol{H_0}$.
  - A measure of fit, the larger the better
    - Large *p-value*: good fit
    - Small *p-value*: poor fit
  - *p-value:* The probability of observing, just by chance, a test statistic as extreme as or even more extreme than what we've actually observed. If $\boldsymbol{H_0}$ is rejected, this ***(p-value)*** is the actual probability that we have made an incorrect decision.

p-Values and "Best Fits"

- Vehicle Arrival Example (cont.):
  - H_0: data is Possion with parameter λ=3.64
  - Test statistics: $\chi_0^2 = 27.68$, with *5* degrees of freedom
  - *p-value = 0.00004*

  ($p\ value = P(\chi_0^2 \geq 27.68) \approx 0.00004$#2

  meaning we would reject H_0 with *0.00004* significance level, hence Poisson is a poor fit.

p-Values and "Best Fits"

- Many software use *p-value* as the ranking measure to automatically determine the "best fit". Things to be cautious about:
  - Software may not know about the physical basis of the data, distribution families it suggests may be inappropriate.
  - Close conformance to the data does not always lead to the most appropriate input model.
  - *p-value* does not say much about where the lack of fit occurs
- Recommended: always inspect the automatic selection using graphical methods.

Fitting a Non-stationary Poisson Process

- Fitting a NSPP to arrival data is difficult, possible approaches:
  - Fit a very flexible model with lots of parameters or
  - Approximate constant arrival rate over some basic interval of time, but vary it from time interval to time interval.

Our focus

- Suppose we need to model arrivals over time [0,T], our approach is the most appropriate when we can:
  - Observe the time period repeatedly and
  - Count arrivals / record arrival times.

Fitting a Non-stationary Poisson Process

- The estimated arrival rate during the *i*th time period is:

$$\hat{\lambda}(t) = \frac{1}{n\Delta t}\sum_{j=1}^{n} C_{ij}$$

where n = # of observation periods, Δt = *time interval length*

C_{ij} = # of arrivals during the ith time interval on the jth observation period

- Example: Divide a *10*-hour business day [*8am,6pm*] into equal intervals $k = 20$ whose length $\Delta t = ½$, and observe over n =3 days

Time Period	Number of Arrivals Day 1	Day 2	Day 3	Estimated Arrival Rate (arrivals/hr)
8:00 - 8:30	12	14	10	24
8:30 - 9:00	23	26	32	54
9:00 - 9:30	27	18	32	52
9:30 - 10:00	20	13	12	30

For instance, 1/3(0.5)*(23+26+32) = 54 arrivals/hour

Selecting Model without Data

- 1st method: If data is not available, some possible sources to obtain information about the process are:
  - **Engineering data:** often product or process has performance ratings provided by the manufacturer or company rules specify time or production standards.
  - **Expert option:** people who are experienced with the process or similar processes, often, they can provide optimistic, pessimistic and most-likely times, and they may know the variability as well.
  - **Physical or conventional limitations:** physical limits on performance, limits or bounds that narrow the range of the input process.
  - **The nature of the process.**
- 2nd method: When data are not available:
  - The uniform, triangular, and beta distributions are often used as input models.

Selecting Model without Data

- Example: Production planning simulation.
  - Input of sales volume of various products is required, salesperson of product XYZ says that:
    - No fewer than *1,000* units and no more than *5,000* units will be sold.
    - Given her experience, she believes there is a *90%* chance of selling more than *2,000* units, a *25%* chance of selling more than *3,500* units, and only a *1%* chance of selling more than *4,500* units.
  - Translating these information into a cumulative probability of being less than or equal to those goals for simulation input:

1	$1000 \le x \le 2000$	0.10
2	$2000 < x \le 3500$	0.75
3	$3500 < x \le 4500$	0.99
4	$4500 < x \le 5000$	1.00

MULTIVARIATE / TIME SERIES

Multivariate and Time-Series Input Models

- Multivariate:
  - For example, lead time and annual demand for an inventory model, increase in demand results in lead time increase, hence variables are dependent.
- Time-series:
  - For example, time between arrivals of orders to buy and sell stocks, buy and sell orders tend to arrive in bursts, hence, times between arrivals are dependent.

Covariance and Correlation

- Consider the model that describes relationship between X_1 and X_2:

$$(X_1 - \mu_1) = \beta(X_2 - \mu_2) + \varepsilon$$

ε is a random variable with mean *0* and is independent of X_2

  - $\beta = 0$, X_1 and X_2 are statistically independent
  - $\beta > 0$, X_1 and X_2 tend to be above or below their means together
  - $\beta < 0$, X_1 and X_2 tend to be on opposite sides of their means

- Covariance between X_1 and X_2 :

$$\text{cov}(X_1, X_2) = E[(X_1 - \mu_1)(X_2 - \mu_2)] = E(X_1 X_2) - \mu_1 \mu_2$$

$$\text{where} \quad cov(X_1, X_2) \begin{cases} = 0, \\ < 0, \\ > 0, \end{cases} \quad \text{then } \beta \begin{cases} = 0 \\ < 0 \\ > 0 \end{cases}$$

Covariance and Correlation

- Correlation between X_1 and X_2 (values between *-1* and *1*):

$$\rho = \text{corr}(X_1, X_2) = \frac{\text{cov}(X_1, X_2)}{\sigma_1 \sigma_2}$$

  - where $corr(X_1, X_2) \begin{cases} = 0, \\ < 0, \\ > 0, \end{cases}$ then $\beta \begin{cases} = 0 \\ < 0 \\ > 0 \end{cases}$
  - The closer ρ is to *-1* or *1*, the stronger the linear relationship is between X_1 and X_2.

Covariance and Correlation

- A time series is a sequence of random variables $X_1, X_2, X_3, \ldots$, are identically distributed (same mean and variance) but dependent.
  - $\text{cov}(X_t, X_{t+h})$ is the *lag-h autocovariance*
  - $\text{corr}(X_t, X_{t+h})$ is the *lag-h autocorrelation*
  - If the autocovariance value depends only on *h* and not on *t*, then the time series is covariance stationary

Multivariate Input Models

- If X_1 and X_2 are normally distributed, dependence between them can be modeled by the bivariate normal distribution with μ_1, μ_2, σ_1^2, σ_2^2 and correlation ρ
  - To Estimate μ_1, μ_2, σ_1^2, σ_2^2, see "Parameter Estimation" (slide 17- 19, or Section 9.3.2 in the textbook)
  - To Estimate ρ, suppose we have n independent and identically distributed pairs (X_{11}, X_{21}), (X_{12}, X_{22}), … (X_{1n}, X_{2n}), then:

$$\hat{\text{cov}}(X_1, X_2) = \frac{1}{n-1}\sum_{j=1}^{n}(X_{1j} - \hat{X}_1)(X_{2j} - \hat{X}_2)$$

$$= \frac{1}{n-1}\left(\sum_{j=1}^{n} X_{1j}X_{2j} - n\hat{X}_1\hat{X}_2\right)$$

$$\hat{\rho} = \frac{\hat{\text{cov}}(X_1, X_2)}{\hat{\sigma}_1\hat{\sigma}_2}$$

Sample standard deviation

Time-Series Input Models

- If $X_1, X_2, X_3, \ldots$ is a sequence of identically distributed, but dependent and covariance-stationary random variables, then we can represent the process as follows:
  - Autoregressive order-1 model, AR(1)
  - Exponential autoregressive order-1 model, EAR(1)
    - Both have the characteristics that:

$$\rho_h = corr(X_t, X_{t+h}) = \rho^h, \quad \text{for } h = 1,2,...$$

    - *Lag-h* autocorrelation decreases geometrically as the lag increases, hence, observations far apart in time are nearly independent

AR(1) Time-Series Input Models

- Consider the time-series model:

$$X_t = \mu + \phi(X_{t-1} - \mu) + \varepsilon_t, \quad \text{for } t = 2,3,...$$

where $\varepsilon_2, \varepsilon_3, \ldots$ are i.i.d. normally distributed with $\mu_\varepsilon = 0$ and variance σ_ε^2

- If X_1 is chosen appropriately, then
  - $X_1, X_2, \ldots$ are normally distributed with *mean* = μ, and *variance* = $\sigma^2/(1-\phi^2)$
  - Autocorrelation $\rho_h = \phi^h$
- To estimate $\phi, \mu, \sigma_\varepsilon^2$:

$$\hat{\mu} = \overline{X}, \qquad \hat{\sigma}_\varepsilon^2 = \hat{\sigma}^2(1-\hat{\phi}^2), \qquad \hat{\phi} = \frac{\hat{\text{cov}}(X_t, X_{t+1})}{\hat{\sigma}^2}$$

where $\hat{\text{cov}}(X_t, X_{t+1})$ is the *lag*-1 autocovariance

EAR(1) Time-Series Input Models

- Consider the time-series model:

$$X_t = \begin{cases} \phi X_{t-1}, & \text{with probability } \phi \\ \phi X_{t-1} + \varepsilon_t, & \text{with probability } 1\text{-}\varphi \end{cases} \qquad \text{for } t = 2,3,...$$

where $\varepsilon_2, \varepsilon_3, \ldots$ are i.i.d. exponentially distributed with $\mu_\varepsilon = 1/\lambda$, and $0 \le \phi < 1$

- If X_1 is chosen appropriately, then
  - $X_1, X_2, \ldots$ are exponentially distributed with *mean* = $1/\lambda$
  - Autocorrelation $\rho_h = \phi^h$, and only positive correlation is allowed.
- To estimate ϕ, λ :

$$\hat{\lambda} = 1/\overline{X}, \qquad \hat{\phi} = \hat{\rho} = \frac{\hat{\text{cov}}(X_t, X_{t+1})}{\hat{\sigma}^2}$$

where $\hat{\text{cov}}(X_t, X_{t+1})$ is the *lag*-1 autocovariance

Summary



- In this lecture, we described the 4 steps in developing input data models:
  - Collecting the raw data
  - Identifying the underlying statistical distribution
  - Estimating the parameters
  - Testing for goodness of fit